بِسْمِ ٱللَّهِ ٱلرَّحْمَٰنِ ٱلرَّحِيمِ

**In the name of Allah, Most Gracious, Most Merciful**

# CSE 4107
# Structured Programming I

Topic: Basic Types

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# Basic Types

`int float, _Bool`: We have already seen them. Its time for the rest of the types

## `Follow the white board`

- *Integer Types:* `int a, b, c;`
- *Floating Types:* `float fa, fb, fc;`
- *Character Types*: `char ch, ca, cb;`

Some other things to discuss:

- Type Conversion
- Type Definitions
- The `sizeof` operator

Source: C Programming Modern Approach Book

# Integer Types

`int`: The whole number.

## Follow the white board

- *Signed Integers*: Left most bit is zero for positive number, one for negative.

```
0111 1111 1111 1111      // 32,767 (2^15 - 1)
```

- *UnSigned Integers:* Left most bit is considered as part of numbers magnitude.

```
1111 1111 1111 1111      // 65,535 (2^16 - 1)
```

- Usually 32 bits, but can be 16 bits in older CPUs
- `long int:`
- `short int:`

Source: C Programming Modern Approach Book

# Integer Types

`int`: The whole number.

`Follow the white board`

| *Type* | *Smallest Value* | *Largest Value* |
|---|---|---|
| `short int` | –32,768 | 32,767 |
| `unsigned short int` | 0 | 65,535 |
| `int` | –32,768 | 32,767 |
| `unsigned int` | 0 | 65,535 |
| `long int` | –2,147,483,648 | 2,147,483,647 |
| `unsigned long int` | 0 | 4,294,967,295 |

- *For 16 bit CPUs*

Source: C Programming Modern Approach Book

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*For 32 bit CPUs*

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| `int` | –2,147,483,648 | 2,147,483,647 |
| `unsigned int` | 0 | 4,294,967,295 |
| `long int` | –9,223,372,036,854,775,808 | 9,223,372,036,854,775,807 |
| `unsigned long int` | 0 | 18,446,744,073,709,551,615 |

*For 64 bit CPUs*

*C99:* `long long int` *and* `unsigned long long` *int are 64 bits*

Source: C Programming Modern Approach Book

# Integer Constant Types

C supports direct use of 3 types of base. Decimal, Octal, Hexadecimal

`Follow the white board`

- *Decimal Integers*: must not start with zero
  int a = 15, b = 255, c = 3278
- *Octal Integers:* must start with zero, digits are between 0 and 7
  int a = 015, b = 0255, c = 0777
- *Hexadecimal Integers:* must start with `0x`, can have digits 0 to 9 and `a` to `f`
  int a = 0x15, b = 0x255, c = 0x7a77f
- `C99: integer constant ends with LL or ll for long long int.`
  `Addin u means unsigned.  long long int a = 10ull`

Source: C Programming Modern Approach Book

# Integer Overflow

Calculated number couldn’t be fitted in the memory.

Follow the white board

```
int a = 1000;
int b = a*a*a*a*a*a*a*a
```

Source: C Programming Modern Approach Book

# Floating Types

The number having fractional parts.

Follow the white board

- *float*: Single precision floating point
- *Double:* Double precision floating point
- *Long double:* Extended-precision floating point
- C99: float, double, long double falls under real floating point, float_complex, double_complex, long double_complex falls under complex types.

- Floating Constant: 57.0 57. 57.0e0 57E0 5.7e1 5.7e+1 .57e2 570.e-1

Source: C Programming Modern Approach Book

# Floating Types

The number having fractional parts.

Follow the white board

| *Type* | *Smallest Positive Value* | *Largest Value* | *Precision* |
|---|---|---|---|
| `float` | $1.17549 \times 10^{-38}$ | $3.40282 \times 10^{38}$ | 6 digits |
| `double` | $2.22507 \times 10^{-308}$ | $1.79769 \times 10^{308}$ | 15 digits |

IEEE Standard

Source: C Programming Modern Approach Book


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THANK
YOU